

Advancing HCI Research and Education within and across South Asia

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ABSTRACT

With the advancements in technologies, the need for Human's perspective and human-centered designs are in much demand and its essential to understand diverse cultural needs. The community in South Asia has been recognized to have a unique and diverse socio-cultural, political, infrastructural, and geographical background of the region. However, we continue to see that the studies presented to the CHI community about South Asia primarily focus on working with and unpacking the regional contextual constraints (of the users and the infrastructures), thus taking a *developmental* stance. We witness a lack of HCI work presented by South Asian research community promoting diverse methods, cultures and behaviors. We believe this is due to limited experience in the field and resources. In this online workshop, we take advanced steps to operationalize collaborations and resource sharing between HCI researchers by presenting their half-baked ideas, presenting published HCI work on other venues, or even sharing challenges faced in a Rejected HCI work in the context of South Asia. Our aim is to broaden the perspective of the CHI research and community towards the contributions from the region *including and beyond development*, by bringing together researchers, designers, and practitioners working or are interested in working within these regions on diverse topics.

KEYWORDS

HCI, South Asia, Beyond development, Vision building

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1 BACKGROUND:

The field of human-computer interaction (HCI) has evolved from a technology-oriented point of view to a human-centered design point of view. Although humankind is the same around the world, there are diverse facets that make communities unique and native. Therefore, HCI researchers and practitioners need to understand that acceptance of technologies [26], expectations [20], and the experience of these communities are highly dependent on contextual factors [4]. We witness communities of research groups emerging, targeting niche groups and for their betterment—For example, ICT for Development (ICT4D) which aims to enhance the quality of life of people originating from or living in “developing countries” [21], HCI Across Borders (HCIxB) that strives for best HCI practices within and across continents [10] and Asian CHI community that focus on Asian sociocultural factors in their research design and implementation of computing systems [29]. Due to the intense of diversity among the cultures, economies and developmental phases of Asia, the expectations to practicality gap within the community is at significant higher level. However, the South Asian continent within Asia is profoundly having similarities and differences in its economies, and sociocultural development. Specifically, researchers from developed countries position the South Asian context uniquely and explore emerging and developing experiences under many dimensions. For example, researchers developed photo-voltaic deformable interfaces using the co-design method in India to

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illustrate the alternative future use of digital technologies [18]. The work situated in the South Asian region is labeled as a contribution from the Global South; thus, the contribution from South Asia has primarily been identified as data points. To expand such a narrow view, a community within the South Asian context formed a formal agenda not only to benefit from the richness and diversity within the society but also to introduce interactions, generate and disseminate knowledge about cultures, theoretical and methodological aspects embodied in living experiences [6]. An open document was curated that discussed about educational technology; publication in HCI; gender, health, and wellbeing and AI for Next Billion.

Since its inception, this will be our formal second gathering to advance and accelerate our approaches to building a strong community within and across researchers interested in contextual work within the South Asian region. We intend to play a catalyst role in bringing HCI educators, researchers, designers, and technology makers at CHI 2023 to the researchers and community in South Asia. Our virtual CHI workshop will provide a platform for researchers to present their half-baked research ideas centered around the South Asian continent, present their work related to South Asia published in other venues, or present their research HCI-related work which got rejected from any top-tier conference venue. By doing that, we aim to listen to the community's needs and challenges and collectively pair to develop solutions. Since the publications and research from South Asia significantly lack in top-tier HCI conferences, we also aim to identify challenges and provide support in knowledge and resources. Thus, our "Advancing HCI Research and Education within and across South Asia" has two main objectives.

- To make South Asia (Global South) a centre stage to make new research perspectives that facilitate goals for the research community.
- To build a platform to share research findings (halfbaked/unpublished/ published/ rejected) centering the South Asian region.
- To explore challenges in HCI-related research and reasons for limited research in this domain originated in the South Asian context.

We plan to explore various themes presented by researchers during the workshop and exchange ideas and facilitate conversations (including anecdotes, experiential information, and field stories). We present a few of the exploratory themes that the organizers have discussed and share such knowledge as panel discussions and invited talks.

1.1 Background to HCI research in South Asia

The scholarship on Human-Computer Interaction (HCI) has seen an increase in the work from South Asia over the years [1]. The growing HCI research work in the area has primarily focused on understanding, designing, and deploying technologies. The wide array of HCI research has focused on Information communication technology (ICT) and its developmental effects on the users, especially related to varied socio-cultural norms, socio-political underpinnings, lack of resources, and income (e.g., [1, 2, 11, 15, 17, 27, 30, 32]). There have also been researched investigations carried out by or in collaboration with researchers from the Global North (e.g.,

[8, 9, 19, 24, 28]) focusing on understanding the challenges and difficulties these under-resourced user groups face. Therefore given the diverse socio-cultural and socio-political fabric of the region and the growing advancements and adoption of technologies, the issues related to accessibility [3], healthcare [12, 31], information literacy, sustainability [5, 25] will continue to pose challenges, increasingly from the perspective of new and emerging technologies [16]. Thus, accounting for the region's diverse offerings to the HCI community and expansion to the HCI scholarship, it becomes essential for the South Asian HCI community to strengthen agendas and share a common vision of HCI by looking at the region as a part of the larger social, cultural and political contexts.

1.2 Background to HCI Education in South Asia

HCI as a discipline is relatively new to South Asia, specifically in the Asian region. South Asia consists of eight countries according to the United Nations and the South Asian Association for Regional Cooperation (SAARC) Council [14]. This includes Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, and Sri Lanka. Asian regions share similarities in terms of diversity with respect to religion, socioeconomic status of people, culture, language, and especially the way academia is structured in terms of schools and colleges [23]. Although prior research has advocated how HCI can have an impact on emerging economies by designing tools and technologies while understanding low-income users, users with limited abilities, or even users under different cultural environments, much of these implementations have been designed with an influence from western cultures. Researchers in South Asia have identified that this may be due to the influence of HCI learning obtained from the west [13, 33]. A multicultural perspective on HCI education is prevalent in HCI education, yet for South Asians "HCI" as a subject has been challenging due to many reasons. Recent research explores what are the challenges and avenues for South Asians to confront HCI as a field to study [7]. Their research found 3 main different avenues for students to explore HCI in South Asia—1) Traditional Avenues such as the usual academic course work and industry workshops, 2) Emergent Avenues such as by the influence of Community Engagement, Social media or Online courses. 3) Inter-sectional Avenues such as through mentoring, similar research papers or even student chapters and groups organized by SIGCHI. While traditional avenues (those primarily constituting institutional offerings) are prevalent, there exist Emergent and Inter-sectional avenues were found to be the contexts where students went beyond institutional boundaries to seek the knowledge they wish to gain. Echoed largely in the study was the lack of HCI coursework within the formal curriculum, thus leading to the creation of emergent avenues by the students who once faced challenges in encountering HCI. As Samson et al. emphasize the importance of fostering good networking among academics and practitioners and growing a research community from Asia, our workshop address the key challenges faced when populating HCI as an academic venue in South Asia [22].

2 ORGANIZERS:

Similar to the past workshop we had in 2021, the organizers bring a wealth of experience in conducting HCI related research while

situating in the continent. At the same time, some of them are HCI educators within South Asian universities, thus empathize the challenges faced by HCI researchers in this context. Although some of the authors were introduced new to the group, their work and the influence they can propagate impact wider audience while situated in South Asia.

- **Dilrukshi Gamage** is a Postdoctoral fellow at the Tokyo Institute of Technology in Japan. Her research interests are intersecting with Computational Social Science and HCI. She is a founding member of the Colombo SIGCHI chapter in Sri Lanka and lead the chapter for two consecutive years while completing her PhD residing in Sri Lanka. She has years of experience organizing CHI workshops, especially HCI Across Borders, and HCI4South Asia together with other organizers in this workshop.
- **Asra Sakeen Wani** is a Ph.D. student in the Dept of Computer Science & Engineering at IIIT-Delhi, India. Her research work lies in the area of Crisis Informatics, Human-Centered Computing (HCC), and Science and Technology Studies (STS).
- **Pranjal Jain** is a PhD Researcher in the Centre for Doctoral Training at Swansea University, focusing on enhancing human interactions and collaborations with intelligence-driven systems. His work involves identifying and capturing tacit knowledge to improve an organisation's knowledge management. His research intersects Computer Science, Neuroscience, Psychology and HCI. He is a founding member of the Srishti SIGCHI chapter in India. He has previously organised CHI and India HCI workshops, especially HCI4SouthAsia and CHIArt, along with other community members.
- **Deepak Ranjan Padhi** is an Assistant Professor at the National Institute of Fashion Technology, Bhubaneswar, India. His primary research interests are Human-Computer Interaction (HCI) and Education Technology (ET). He has five years of experience organizing workshops and courses at various avenues.
- **Pranjal Protim Borah** is a Ph.D. student working in accessibility and tangible interaction in the Department of Design, IIT Guwahati, India. His research interests include designing inclusive interaction techniques for emerging technologies.
- **Ambika R Menon** is a recent undergraduate from Srishti Institute of Art, Design, and Technology. Currently engaging as an affiliated undergraduate research assistant at UXP2 Lab, Purdue University exploring design through the lens of ethics and social responsibility. As a human-centered designer in the making, her interests and strengths lie at the intersection of User experience research, Systems thinking practice, and Digital product design.
- **Wricha Mishra** is an Associate Professor in the Department of User Experience Design at MITID, MIT ADT University Pune, India. Her research interest revolves around Conversational Interfaces, Human Factors and Inclusive Design.
- **Muneeb I. Ahmad** is a lecturer in Computer Science at Swansea University in the United Kingdom. His research focuses on employing methods from the field of human-centered design to create intelligent interactive systems such

as social robots, and user interfaces that can adapt to users through real-time analysis of their social signals and physiological data.

- **Nova Ahmed** is a Professor, at the Department of Electrical and Computer Engineering, North South University, Bangladesh. Her research extensively looked at marginalized groups in Bangladesh and South Asian regions and embedded socio-techno-cultural perspectives. She is playing a key role in SIGCHI Dhaka and has taken many initiatives to populate HCI in the region through workshops and summer schools.
- **Sayan Sarcar** is a Lecturer in Human-Computer Interaction within the School of Computing and Digital Technology at Birmingham City University, England. His research sits at the intersection of Human-Computer Interaction (HCI) and Accessibility. His research emphasizes improving human individual abilities through developing intelligent systems using design and modeling practices, specifically focused on individual differences in users' sensorimotor abilities.
- **Suleman Shahid** is an Assistant Professor in Computer Science at Lahore University of Management Science (LUMS) and Director of the LUMS Learning Institute. At LUMS, he also directs the Computer-Human Interaction and Social Experience Lab (CHISEL). His research focuses at the intersection of design, technology and well-being with applications to assistive technologies to enhance the quality of life of persons with disabilities, and persons with mental health conditions, learning technologies for children, and social computing.
- **Devanuj Kanta Balkrishan** is an Associate Professor in Institute of Design at JK Lakshmipat University, India. A designer with qualifications in three different domains of design namely, Architecture from IIT, Roorkee, Product Design from IIT, Delhi and Human-Computer Interaction from IIT, Bombay. He is trained to work on human problems related to habitat or usability.

3 PRE-WORKSHOP PLANS AND WEBSITE:

We will invite applications from researchers, designers, practitioners, and technology makers, including undergraduate and early career researchers working in the context of South Asia. The workshop will be conducted online. We will use the Zoom platform for the interactions. The presentations will be shared by the participants of the workshop and a larger community of South Asian researchers. The repositories, in the form of video-recordings and documents, will be preserved and hosted on our website for a wider accessibility. All relevant information about the workshop will be updated and preserved on the website. Participants will be invited to present or take part in the workshop in 4 distinct types. To include individuals with limited experience in writing scientific articles and presenting research, we intend to invite participants based on self-nominations or nominated through a South Asian researcher. Details of the participation type and selection criteria are described below.

3.1 Participation Types

- **Presenter of a half-baked idea:** Interested presenters will be required to submit approximately 500 words of abstract and the idea's title. The abstract should cover a research gap or problem statement they would like to address and potential solutions (not necessarily concrete). Their idea should be aligned with the visions and missions of the HCI4South Asia group within the field of HCI.
- **Presenter of a previously published HCI paper:** We will invite participants who have published during 2021–2022 in HCI conferences or journals to share their research findings, methods and challenges faced conducting the research. We will request them to share a draft version of their paper to share with the participants. This will provide an opportunity for other participants to learn about the success and failures of conducting HCI research. The theme, methods, or findings of the research should be relevant to the South Asian context.
- **Presenter of a submitted/rejected HCI paper:** We will invite participants who have submitted a manuscript to an HCI venue (conference or journal), and the manuscript is under review or got a rejection for the same. We invite them to share experiences of what went wrong or could have succeeded. The presenters will be required to submit approximately 500 words of abstract of their paper. This abstract will be shared with the participants, and the paper should be in HCI and the South Asian context.
- **Participants as self-nominated or nominated by a South Asian Researcher:** In the past, we have experienced that many of the students in the South Asian region are not aware of CHI workshops, nor do they have experience writing papers or engaging in critical discussion for HCI. We will open a specific type of submission where a South Asian researcher can nominate a less experienced (ideally students) to join the workshop without any particular artefact, observe the conversations and presentations, and engage with researchers whenever they feel comfortable. At the same time, this nomination will be further expanded to self-nominations, widening every opportunity to attract early researchers.

3.2 Selection Criteria

Each submission will be selected through evaluation by a small jury and the Workshop organizing committee. Selections will be based on the contributions and implications to South Asian context but not limited to those.

4 WEBSITE AND TWITTER:

Details about the workshop, including key dates, will be available via the workshop website at <http://hci4south.asia> and twitter <https://twitter.com/HCI4SouthAsia>. This website and Twitter handle will act as the information catalyst in the weeks leading up to the workshop. It would also act as a repository for participants' submissions, and participants will be encouraged to familiarize themselves with the other attendees before the day of the workshop.

5 WORKSHOP STRUCTURE:

The workshop will be held as a **half-day, fully virtual workshop** comprising three sessions expanding for five-hours (see Table 1). We expect to have an opening keynote and a closing keynote from a well-known researcher who has influenced and impacted many in the region, presentations from the participants who got their submissions accepted by our workshop organizers, panel discussions and group discussions based on the themes of the presentations. After the decision, we will post the final plan and the schedule on the website.

We will start with an opening presentation introducing our community and a summary of the previous workshop in 2021 at CHI. We will highlight the plans for advancing our collaborative work. Then our opening keynote will be presented with questions and answers. Our panel members will discuss challenges and opportunities in creating borders and bridges at career, geographies, and internet access levels. We expect the panel to introduce key themes (e.g., internet access issues, geographical challenges, improved access).

We will split into smaller breakout rooms following a short break based on key themes for our subsequent sessions. In our breakout rooms, we will have discussion sessions where each participant will discuss their ideas in the context of the theme. We expect members to discuss the posters and for groups to reflect on emerging insights and solutions.

We will regroup in the final session to share our key insights from the breakouts after another break. We will spend the second half of the last session collectively synthesizing ideas into themes to consider for the sustainability of an increasingly virtual/hybrid community (e.g., *What do various borders in HCI look like?*, *What can we do about them?*). A final report will be shared as a blog post to continue the conversation with the HCI4SouthAsia community.

Time	Activity
10 mins	Welcome from the organizers
20 mins	Opening Keynote
5 mins	Break: <i>optional—social breakout rooms</i>
30 mins	Participants presentations
25 mins	Feedback and discussions in themed breakout rooms
15 mins	Break: <i>optional—social breakout rooms</i>
20 mins	Closing Keynote
20 mins	Reflections and synthesis

Table 1: Proposed Agenda for HCI4SouthAsia 2023

6 POST-WORKSHOP PLANS:

We aim to achieve a three-fold outcome from the workshop:

(1) **South Asian Community Creation:** First, we hope to identify and create a community of HCI researchers, designers, practitioners, and technology makers engaged in work with and within South Asia. Since this will be our second workshop at CHI, we will debrief our collective vision of the community about their future agendas, aspirations, and challenges *beyond development* work. The participants will also be invited to join our mailing list to receive updates about upcoming events and initiatives. The identified vital themes from the workshop discussion, brainstorming, and design

activity will be available in the form of the living document for further building and improving by the community.

(2) Broader HCI Community Discussion and Collaborations: Second, the half baked ideas submitted to the workshop will ignite potential collaborations and invite expertise knowledge. Our workshop will play a catalysis role in connecting collaborations. At the same time our workshop conversations will result in initiating an HCI community-wide discussion reflecting plans and agendas suggested and shaped by our participants, representing the South Asian HCI community. This community discussion will contribute in (re)shaping our future workshop agendas and sustaining the ongoing conversation beyond the workshop.

(3) Publication(s): Finally, we have strong aims to produce one or more publication(s) to present new themes, principles, and visions for conducting human-centered research and design in South Asia. The publication venue(s) will be determined according to the number of generated outcomes (themes, principles, etc.): a shorter list may result in an article for *Communications of the ACM* or *ACM Interactions*; or other related venues. Depending on the workshop outcome, workshop participants will either be acknowledged for their contribution to this publication(s), or be invited for co-authorships. Participants will also be strongly encouraged to publish the results of their discussions with others they have connected with through the workshop. The organizers will as well work with interested participants from the South Asian region to provide writing support as required.

Besides these three aims, we will also identify additional pathways to outputs that may benefit participants directly, in particular those that may have an impact beyond the CHI community. Participants will be asked to report any resulting publications or other outcomes so that the organizers can send relevant updates.

7 CALL FOR PARTICIPATION:

We aim to continue building "HCI4SouthAsia", a strong community of HCI researchers and practitioners who work/will work in the South Asian region (i.e., in and around the countries Afghanistan, Bangladesh, Bhutan, India, Maldives, Nepal, Pakistan, and Sri Lanka). We are organizing a half-day online workshop at CHI 2023, titled: **"Advancing HCI Research and Education within and across South Asia"**.

The primary objective of the workshop is to provide an international platform to showcase – to the "center" – the research (including and beyond development), challenges, experiential information, and field stories, from (marginal) communities across the South Asian Region, and help to collectively build a vision for the community. We invite participants to join our remotely using a common digital medium. The half-day workshop will include opportunities to present your research, research ideas and collaborate and also to share the challenges of publishing research papers in the field of HCI in the global south context.

Interested participants are invited to fill out a submission link and share details about their published research, methods or preliminary ideas for potential research and their aspirations to find collaborative partners from the workshop. Not only published research or ideas, but we also open up submissions as nominations to

take part in the workshop as a passive listeners. Interested participants may self-nominate or South Asian researchers may encourage their participants can submit a detailed write-up about their research experiences and field stories related to HCI research or about designing/deploying technologies in the region. colleagues and students who have less exposure to the HCI should nominate. Participants will be accepted based on how their responses align with the workshop theme. The last day to apply is **February 21, 2023**. More details about the workshop and its organizers can be found on the workshop website at <http://hci4south.asia> and <https://twitter.com/HCI4SouthAsia>.

REFERENCES

- [1] Nicola Dell and Neha Kumar. 2016. The Ins and Outs of HCI for Development. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 2220–2232. <https://doi.org/10.1145/2858036.2858081>
- [2] Devanuj Kanta Balkrishnan and Anirudha Joshi. 2013. Technology Adoption by 'emergent' Users: The User-Usage Model. In *Proceedings of the 11th Asia Pacific Conference on Computer Human Interaction* (Bangalore, India) (APCHI '13). Association for Computing Machinery, New York, NY, USA, 28–38. <https://doi.org/10.1145/2525194.2525209>
- [3] Anita Ghai, Vidhya Yella Reddy, and M Swaminathan. 2020. Assistive technology research and disability studies in the global South: the need for synergy.
- [4] Gillian R Hayes. 2020. Inclusive and engaged HCI. *Interactions* 27, 2 (2020), 26–31.
- [5] Mohit Jain, Ankit Agrawal, Sunil K. Ghai, Khai N. Truong, and Deva P. Seetharam. 2013. "We Are Not in the Loop": Resource Wastage and Conservation Attitude of Employees in Indian Workplace. In *Proceedings of the 2013 ACM International Joint Conference on Pervasive and Ubiquitous Computing* (Zurich, Switzerland) (UbiComp '13). Association for Computing Machinery, New York, NY, USA, 687–696. <https://doi.org/10.1145/2493432.2493444>
- [6] Pranjal Jain, Samia Ibtasam, Sumita Sharma, Nilavra Bhattacharya, Anupriya Tuli, Dilrukshi Gamage, Dhruv Jain, Rucha Tulaskar, Priyank Chandra, Lubna Razaq, et al. 2021. From the Margins to the Centre: Defining New Mission and Vision for HCI Research in South Asia. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–6.
- [7] Pranjal Jain, Anirudh Nagraj, Kartik Joshi, Taru Jain, Dilrukshi Gamage, Sayan Sarcar, and Nova Ahmed. 2022. HCI Knowledge Dissemination in South Asia through both Coursework and Community Engagement. In *Proceedings of the 2021 EduCHI Symposium at the CHI Conference on Human Factors in Computing Systems*.
- [8] Matthew Kam, Aishvarya Agarwal, Anuj Kumar, Siddhartha Lal, Akhil Mathur, Anuj Tewari, and John Canny. 2008. Designing E-Learning Games for Rural Children in India: A Format for Balancing Learning with Fun. In *Proceedings of the 7th ACM Conference on Designing Interactive Systems* (Cape Town, South Africa) (DIS '08). Association for Computing Machinery, New York, NY, USA, 58–67. <https://doi.org/10.1145/1394445.1394452>
- [9] Konstantinos Kazakos, Siddhartha Asthana, Madeline Balaam, Mona Dugal, Amey Holden, Limalemla Jamir, Nanda Kishore Kannuri, Saurabh Kumar, Amarendar Reddy Manindla, Subhashini Arcot Manikam, GVS Murthy, Papreen Nahar, Peter Phillimore, Shreyaswii Sathyanath, Pushpendra Singh, Meenu Singh, Pete Wright, Deepika Yadav, and Patrick Olivier. 2016. A Real-Time IVR Platform for Community Radio. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 343–354. <https://doi.org/10.1145/2858036.2858585>
- [10] Neha Kumar, Vikram Kamath Cannanure, Dilrukshi Gamage, Annu Sible Prabhakar, Christian Sturm, Cuauhtémoc Rivera Loaiza, Dina Sabie, Md Moinuddin Bhuiyan, and Mario A Moreno Rocha. 2020. HCI across borders and sustainable development goals. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–8.
- [11] Neha Kumar, Naveena Karusala, Azra Ismail, and Anupriya Tuli. 2020. Taking the Long, Holistic, and Intersectional View to Women's Wellbeing. *ACM Trans. Comput.-Hum. Interact.* 27, 4, Article 23 (jul 2020), 32 pages. <https://doi.org/10.1145/3397159>
- [12] Eric C. Larson, Mayank Goel, Morgan Redfield, Gaetano Boriello, Margaret Rosenfeld, and Shwetak N. Patel. 2013. Tracking Lung Function on Any Phone. In *Proceedings of the 3rd ACM Symposium on Computing for Development* (Bangalore, India) (ACM DEV '13). Association for Computing Machinery, New York, NY, USA, 29, 2 pages. <https://doi.org/10.1145/2442882.2442917>
- [13] Shaimaa Lazem. 2021. HCI education of choice: on becoming critical and growing inclusivity. *XRDS: Crossroads, The ACM Magazine for Students* 27, 3 (2021), 46–49.

- [14] World Bank Library. 2014. South Asia Report by World Bank.
- [15] Indrani Medhi Thies. 2017. SIGCHI Social Impact Award Talk – Designing for Low-Literate Users. In *Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI EA '17). Association for Computing Machinery, New York, NY, USA, 8–9. <https://doi.org/10.1145/3027063.3058589>
- [16] Indrani Medhi Thies, Nandita Menon, Sneha Magapu, Manisha Subramony, and Jacki O'Neill. 2017. How Do You Want Your Chatbot? An Exploratory Wizard-of-Oz Study with Young, Urban Indians. In *Human-Computer Interaction - INTERACT 2017*, Regina Bernhaupt, Girish Dalvi, Anirudha Joshi, Devanuj Kanta Balkrishnan, Jacki O'Neill, and Marco Winckler (Eds.). Springer International Publishing, Cham, 441–459.
- [17] Deepak Ranjan Padhi, Anirudha Joshi, Abhishek Shrivastava, and Rucha Tulaskar. 2018. Hierarchy or List? Comparing Menu Navigation by Emergent Users. In *Proceedings of the 9th Indian Conference on Human Computer Interaction* (Bangalore, India) (IndiaHCI'18). Association for Computing Machinery, New York, NY, USA, 29–34. <https://doi.org/10.1145/3297121.3297125>
- [18] Dani Kalarikalayil Raju, Krishna Seunarine, Thomas Reitmaier, Gethin Thomas, Yogesh Kumar Meena, Chi Zhang, Adam Pockett, Jennifer Pearson, Simon Robinson, Matt Carnie, Deepak Ranjan Sahoo, and Matt Jones. 2021. PV-Pix: Slum Community Co-Design of Self-Powered Deformable Smart Messaging Materials. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 304, 14 pages. <https://doi.org/10.1145/3411764.3445661>
- [19] Divya Ramachandran, John Canny, Prabhu Dutta Das, and Edward Cutrell. 2010. Mobile-Izing Health Workers in Rural India. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Atlanta, Georgia, USA) (CHI '10). Association for Computing Machinery, New York, NY, USA, 1889–1898. <https://doi.org/10.1145/1753326.1753610>
- [20] Katharina Reinecke and Krzysztof Z. Gajos. 2014. Quantifying visual preferences around the world. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 11–20.
- [21] Nithya Sambasivan and Thomas Smyth. 2010. The human infrastructure of ICTD. In *Proceedings of the 4th ACM/IEEE international conference on information and communication technologies and development*. 1–9.
- [22] Briane Paul V. Samson, Suleman Shahid, Akihiro Matsufuji, Chat Wacharamanatham, Toni-Jan Keith P. Monserrat, Keyur Sorathia, Masitah Ghazali, Shio Miyafuji, Nova Ahmed, Kazuyuki Fujita, et al. 2020. Asian CHI symposium: HCI research from asia and on asian contexts and cultures. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–6.
- [23] Nasra M Shah. 1994. An overview of present and future emigration dynamics in South Asia. *International migration (Geneva, Switzerland)* 32, 2 (1994), 217–268.
- [24] Yedendra B. Shrinivasan, Mohit Jain, Deva P. Seetharam, Abhishek Choudhary, Elaine M. Huang, Tawanna Dillahunt, and Jennifer Mankoff. 2013. Deep Conservation in Urban India and Its Implications for the Design of Conservation Technologies. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Paris, France) (CHI '13). Association for Computing Machinery, New York, NY, USA, 1969–1978. <https://doi.org/10.1145/2470654.2466261>
- [25] Yedendra B. Shrinivasan, Mohit Jain, Deva P. Seetharam, Abhishek Choudhary, Elaine M. Huang, Tawanna Dillahunt, and Jennifer Mankoff. 2013. Deep Conservation in Urban India and Its Implications for the Design of Conservation Technologies. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Paris, France) (CHI '13). Association for Computing Machinery, New York, NY, USA, 1969–1978.
- [26] Detmar Straub, Mark Keil, and Walter Brenner. 1997. Testing the technology acceptance model across cultures: A three country study. *Information & management* 33, 1 (1997), 1–11.
- [27] Sharifa Sultana, Syed Ishtiaque Ahmed, and Susan R. Fussell. 2019. "Parar-Daktar Understands My Problems Better": Disentangling the Challenges to Designing Better Access to Healthcare in Rural Bangladesh. *Proc. ACM Hum.-Comput. Interact.* 3, CSCW, Article 168 (nov 2019), 27 pages. <https://doi.org/10.1145/3359270>
- [28] Sharifa Sultana, François Guimbretière, Phoebe Sengers, and Nicola Dell. 2018. Design Within a Patriarchal Society: Opportunities and Challenges in Designing for Rural Women in Bangladesh. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3173574.3174110>
- [29] Josh Tedjasaputra, Briane Paul V. Samson, Masitah Ghazali, Eunice Sari, Sayan Sarcar, Dilrukshi Gamage, Kazuyuki Fujita, Pranjal Jain, Amit Jena, Toni-Jan Keith Palma Monserrat, et al. 2021. Asian CHI Symposium: HCI Research from Asia and on Asian Contexts and Cultures. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–5.
- [30] Anupriya Tuli, Shruti Dalvi, Neha Kumar, and Pushpendra Singh. 2019. "It's a Girl Thing": Examining Challenges and Opportunities around Menstrual Health Education in India. *ACM Trans. Comput.-Hum. Interact.* 26, 5, Article 29 (jul 2019), 24 pages. <https://doi.org/10.1145/3325282>
- [31] Edward Jay Wang, Junyi Zhu, Mohit Jain, Tien-Jui Lee, Elliot Saba, Lama Nachman, and Shwetak N. Patel. 2018. Seismo: Blood Pressure Monitoring Using Built-in Smartphone Accelerometer and Camera. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–9. <https://doi.org/10.1145/3173574.3173999>
- [32] Asra Sakeen Wani, Divyanshu Kumar Singh, and Pushpendra Singh. 2022. "Hartal (Strike) Happens Here Everyday": Understanding Impact of Disruption on Education in Kashmir. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems* (New Orleans, LA, USA) (CHI '22). Association for Computing Machinery, New York, NY, USA, Article 315, 17 pages. <https://doi.org/10.1145/3491102.3502126>
- [33] Marisol Wong-Villacres, Adriana Alvarado Garcia, and Javier Tibau. 2020. Reflections from the classroom and beyond: Imagining a decolonized hci education. In *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–14.